SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3– Output Prediction Questions

Course Code:	CSA0303	**Course Title:**	Data Structures
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – Debugging and Optimization Questions
Weightage:	10% of CIA	**Total Marks:**	100
CO2 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	M.NANDHINI	**Reg. No.:**	192472318
Section / Batch:	SLOT-C	**Date:**	30-07-26

Code of Conduct

I _____M.NANDHINI(192472318)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____M.NANDHINI_____

Instructions

- This test contains 10 MCQ Based Questions. Total: 100 Marks.
- When a student answer using saveetha compiler, execute the code, Test the test cases , While evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are permitted in the class. Take it as test and submit in the saveetha lab compiler class.

Annexure A – Output Prediction Questions

1.Write a C program that takes an array as input and returns it reversed.

2.Write a C program for palindrome

3.Write a C program for Searching an Element in a Sorted Array

4.Write a C program for Count the Occurrences of an Element

5.Write a program for binary search

6.Write a C program for Merge Two Arrays

7.Write a C program for Find the Largest and Smallest Element in an Array

8.Write a C program for Find the Second Largest Element

9.Write a C program for Find the First and Last Occurrence of an Element

10.Write a C program for Find Missing Element in an Array

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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Student 20 **MANDEM NANDHINI** 192472318RC 4 / 10 submitted

Q1. Reverse a list

— Not Submitted —

Q2. Reverse a string

CODE

```
#include<stdio.h>
int main(){
    char s[20];
    int i=0,j=0,f=1;
    scanf("%s",s);
    while(s[j]≠'\0')
        j++;
    j--;
    while(i<j)
        if(s[i]≠s[j]){
            f=0;
            break;
        }
    if(f)
        printf("palindrom");
    else
        printf("not a palindrom");
    return 0;

}
```

INPUT

```
hello
```

OUTPUT

```
(no output)
```

Q3. Searching

CODE

```
#include<stdio.h>
int main(){
    int a[]={1,2,3,4,5},x=3,i;
    for(i=0;i<5;i++)
        if(a[i]==x){
            printf("found");
            return 0;
        }
    printf("not found");
}
```

OUTPUT

```
(no output)
```

Q4. Searching

— Not Submitted —

Q5. Array

— Not Submitted —

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Q6. Merging

CODE

```
#include<stdio.h>
int main(){
    int a[10],b[10],c[20],n1,n2,i;
    scanf("%d",&n1);
    for(i=0;i<n1;i++)
        scanf("%d",&a[i]);
    scanf("%d",&n2);
    for(i=0;i<n2;i++)
        scanf("%d",&b[i]);
    for(i=0;i<n1;i++)
        c[i]=a[i];
    for(i=0;i<n2;i++)
        c[n1+i]=b[i];
    for(i=0;i<n1+n2;i++)
        printf("%d",c[i]);
    return 0;

}
```

INPUT

```
2
4 5
3
6 7 8
```

OUTPUT

```
(no output)
```

Q7. Largest and Smallest element

— Not Submitted —

Q8. Second Largest Element

CODE

```
#include<stdio.h>
int main(){
    int a[10],i,n;
    int lar,slar;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    lar=slar=a[0];
    for(i=1;i<n;i++){
        if(a[i]>lar){
            slar=lar;
            lar=a[i];
        }
        else
            if(a[i]>slar && a[i]≠lar)
                slar=a[i];
    }
    printf("%d",slar);
    return 0;
}
```

INPUT

```
4
12 35 55 21
```

OUTPUT

```
35
```

Q9. Occurrence of an elemnet

— Not Submitted —

Q10. Missing Element

— Not Submitted —